Cross-Validated Gradient-Boosted Risk Stratification with Engineered Features and SHAP Explainability for Mortality Prediction in Heart Failure

Abstract

Background. Early identification of patients with heart failure who are at high risk of death allows clinical teams to concentrate monitoring and treatment where it is most needed. A model intended for this purpose must discriminate well, produce probabilities that match observed event frequencies, and be explainable at the level of the individual patient.

Methods. The Heart Failure Clinical Records dataset (299 patients, 96 deaths) was used, with the follow-up time excluded from the predictors to prevent target leakage. In a first stage, a standardised logistic regression and a LightGBM classifier were compared on a stratified 80/20 hold-out split, the LightGBM model was calibrated by Platt scaling, and it was explained with SHAP. In a second stage, three derived features (the ratio of serum creatinine to ejection fraction, the ratio of serum creatinine to serum sodium, and an age-by-hypertension interaction) were added, and LightGBM, random forest, CatBoost and a soft-voting ensemble were compared by 5-fold stratified cross-validation. CatBoost was fitted on all patients and explained with SHAP.

Results. On the hold-out set (n = 60), the ROC-AUC was 0.734 for logistic regression and 0.759 for LightGBM, the PR-AUC was 0.591 and 0.542, and Platt scaling lowered the LightGBM Brier score from 0.192 to 0.186. Under cross-validation the four models reached mean ROC-AUC values of 0.784 to 0.798, PR-AUC values of 0.662 to 0.676 and Brier scores of 0.174 to 0.178; all differences between models were smaller than the standard deviation across folds (0.060 to 0.075 for ROC-AUC). SHAP attribution ranked serum creatinine, ejection fraction and age highest in the first stage and the creatinine-to-ejection-fraction ratio, age and ejection fraction highest in the second, with the direction of every main effect agreeing with clinical expectation.

Conclusions. Shallow tree ensembles on a small set of routine measurements gave moderate, clinically coherent risk estimates. The pre-specified targets of ROC-AUC ⩾ 0.85 and Brier score ⩽ 0.15 were not reached in either stage, and the small cohort limits what can be concluded about the relative merit of the models. External validation, calibration of the final model and systematic tuning are required before any use in decision support.

Keywords: heart failure; mortality risk prediction; gradient boosting; CatBoost; LightGBM; probability calibration; SHAP; explainable machine learning; tabular clinical data

1. Introduction

Heart failure is a chronic syndrome in which the heart cannot supply the body with adequate blood flow, and patients with an established diagnosis face a substantial and heterogeneous risk of death. Because clinical attention, follow-up capacity and treatment intensity are limited, a practical question is which patients should be considered at highest risk on the basis of information that is already collected at the bedside or in routine laboratory work. Statistical and machine-learning models that map such measurements to an individual risk estimate are a natural tool for this purpose.

For a risk model to be useful in a clinical setting, accuracy alone is not sufficient. Three properties are usually required. The first is discrimination: the model should rank patients who will die above those who will not. The second is calibration: a predicted risk of 0.30 should correspond to an observed event frequency close to 30%, since clinicians act on the probability and not only on the ranking. The third is explainability: the contribution of each measurement to a given patient's score should be visible, so that the estimate can be checked against clinical judgement. When the outcome is imbalanced, as it is here, and when a missed high-risk patient is costlier than a false alarm, threshold-free metrics that are sensitive to the minority class, such as the area under the precision-recall curve, are preferred over raw accuracy.

The data used in this work derive from the clinical records of patients with heart failure first described by Ahmad et al. (2017). Chicco and Jurman (2020) showed that survival in this cohort can be predicted to a useful degree from serum creatinine and ejection fraction alone, which suggests that a compact and interpretable model is a realistic aim. The cohort is small, however, and any conclusions about model ranking or about absolute performance must be drawn with corresponding caution.

The work reported here was carried out in two stages on this dataset. The first stage builds a leakage-free baseline, compares a linear model with gradient boosting, calibrates the better of the two and explains it. The second stage asks whether three clinically motivated derived features, a wider set of model families and a more stable evaluation protocol improve on the baseline. The specific objectives were:

1. to construct a predictive pipeline that excludes variables not available at the time a prediction is required;
2. to compare a regularised linear baseline with gradient-boosted trees under a stratified hold-out protocol;

3. to assess whether engineered ratio and interaction features and ensemble methods improve performance under 5-fold stratified cross-validation;
4. to evaluate the calibration of predicted probabilities and the effect of Platt scaling;
5. to explain model behaviour globally and for individual patients with SHAP values; and
6. to compare the results with the performance targets set for the project: ROC-AUC of at least 0.85 and Brier score of at most 0.15, with PR-AUC reported as the primary measure for the minority class.

The remainder of the paper describes the dataset (Section 2), the methods of each stage (Section 3), the results (Sections 4 and 5), and a discussion of the findings, limitations and conclusions (Sections 6 to 8).

2. Dataset and exploratory analysis

2.1 Source and cohort

The analysis uses the Heart Failure Clinical Records dataset distributed on Kaggle, which comes from the retrospective study of Ahmad et al. (2017) and was later analysed by Chicco and Jurman (2020). It comprises 299 patients aged 40 to 95 years, of whom 194 are men and 105 are women, and it was collected at a single cardiology institute in Faisalabad, Pakistan. Each patient is described by 11 clinical variables, the follow-up period in days, and a binary indicator of death during follow-up. The dataset was chosen from the two options recommended for the project because it is clean, tabular and small enough to be processed on a laptop CPU in seconds.

The file contains 13 columns and no missing values, so no imputation was necessary. The five binary variables (anaemia, diabetes, high blood pressure, sex and smoking) are already coded as 0/1, and no categorical encoding was required. The six continuous variables have between 17 and 208 distinct values, which confirms that they are measured quantities and not coded categories.

2.2 Variables

Variable	Description (unit)	Mean (SD) or n (%)	Median [min, max]
age	Age of the patient (years)	60.83 (11.89)	60 [40, 95]
anaemia	Decrease of red blood cells or haemoglobin (0/1)	129 (43.1%)	not applicable

Variable	Description (unit)	Mean (SD) or n (%)	Median [min, max]
creatinine_phosphokinase	Level of the CPK enzyme in the blood (mcg/L)	581.84 (970.29)	250 [23, 7,861]
diabetes	Patient has diabetes (0/1)	125 (41.8%)	not applicable
ejection_fraction	Percentage of blood leaving the heart at each contraction (%)	38.08 (11.83)	38 [14, 80]
high_blood_pressure	Patient has hypertension (0/1)	105 (35.1%)	not applicable
platelets	Platelet count in the blood (kiloplatelets/mL)	263,358 (97,804)	262,000 [25,100, 850,000]
serum_creatinine	Level of creatinine in the blood (mg/dL)	1.39 (1.03)	1.1 [0.5, 9.4]
serum_sodium	Level of sodium in the blood (mEq/L)	136.63 (4.41)	137 [113, 148]
sex	Sex coded 1 (men)	194 (64.9%)	not applicable
smoking	Patient smokes (0/1)	96 (32.1%)	not applicable
time	Follow-up period (days); excluded from modeling	130.26 (77.61)	115 [4, 285]
DEATH_EVENT	Death during follow-up (outcome)	96 (32.1%)	not applicable

Table 1. Description and summary statistics of the 13 variables (N = 299).

2.3 Outcome distribution

Of the 299 patients, 96 (32.1%) died during follow-up and 203 (67.9%) survived, so the classes are moderately imbalanced. A classifier that ignores the predictors would obtain a PR-AUC equal to the prevalence, approximately 0.32, and a Brier score of approximately 0.218, which is the product of the two class proportions. These two values are used throughout as no-skill reference points.

2.4 Exploratory analysis

Patients who died were older, had a lower ejection fraction, a higher serum creatinine and slightly lower serum sodium and platelet counts than survivors (Table 2).

Variable	Survived (n = 203)	Died (n = 96)	Difference (died minus survived)
Age (years)	58.76	65.22	+6.46
Creatinine phosphokinase (mcg/L)	540.05	670.20	+130.15
Ejection fraction (%)	40.27	33.47	−6.80
Platelets (kiloplatelets/mL)	266,657	256,381	−10,276
Serum creatinine (mg/dL)	1.18	1.84	+0.66
Serum sodium (mEq/L)	137.22	135.38	−1.84

Table 2. Mean value of the continuous predictors by outcome.

The linear correlations of the variables with the outcome are modest (Figure 1). Excluding follow-up time, the largest in absolute value are serum creatinine (+0.29), ejection fraction (−0.27), age (+0.25) and serum sodium (−0.20). High blood pressure, anaemia and creatinine phosphokinase are weakly positive (+0.06 to +0.08), platelets weakly negative (−0.05), and diabetes, sex and smoking are essentially uncorrelated ($|r| \leqslant 0.01$). Follow-up time has the strongest association (−0.53), but it is excluded from the models for the reason given in Section 2.5.



Figure 1. Pearson correlation of each variable with DEATH_EVENT, ordered from the most positive to the most negative.

The distributions of the continuous variables by outcome (Figure 2) show the same pattern. The median age is about 60 years among survivors and 65 among patients who died, the median ejection fraction is about 38% against 30%, and the median serum sodium is about 137 against 135.5 mEq/L. Serum creatinine is concentrated near 1 mg/dL in survivors, whereas the distribution among deaths is shifted upward and has a long tail reaching 9.4 mg/dL. Platelet counts and creatinine phosphokinase overlap almost completely between the groups. Creatinine phosphokinase, serum creatinine and platelets are strongly right-skewed with several extreme values (for example a CPK maximum of 7,861 against a median of 250), which is of little consequence for tree-based models, since these are invariant to monotone transformations, but relevant to the logistic regression baseline.

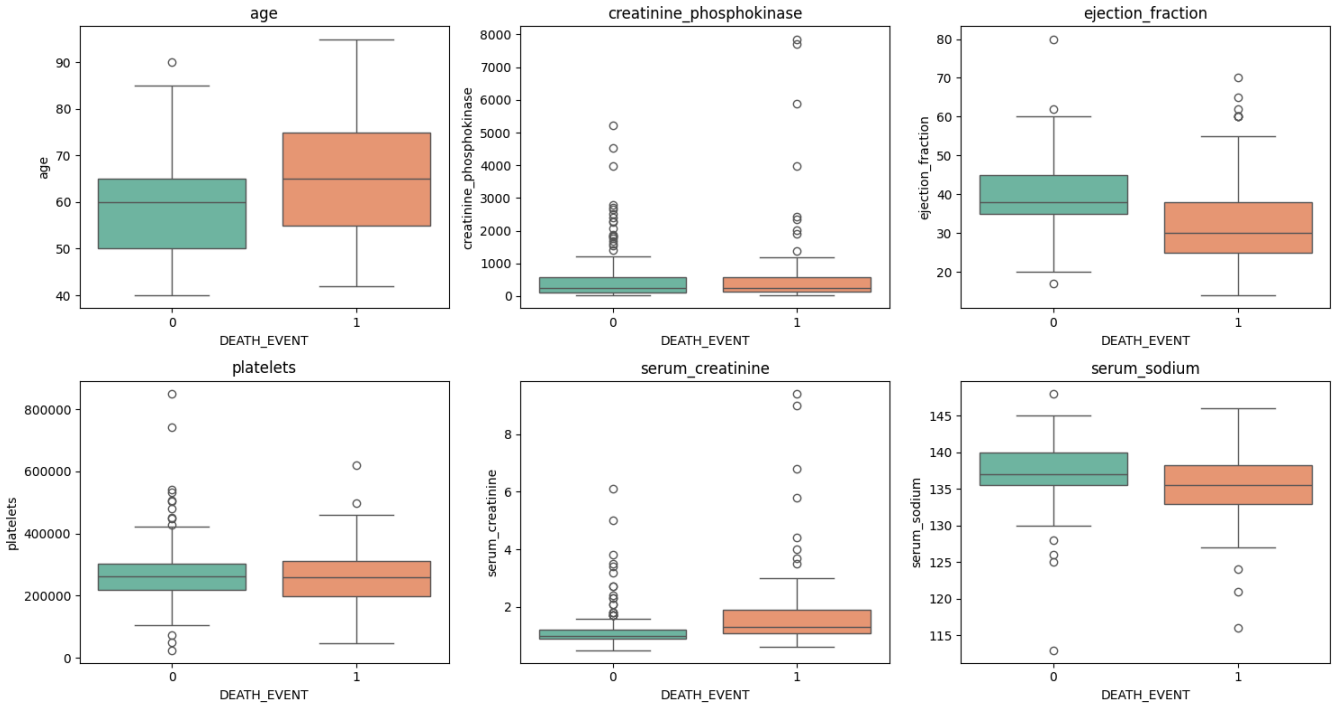


Figure 2. Box plots of the six continuous predictors for survivors (0) and patients who died (1).

2.5 Exclusion of the follow-up variable

The variable `time` has the strongest association with the outcome but was removed from the feature set. It records the length of follow-up, which is itself shortened by death, and it would not be known at the moment a risk estimate is required. Retaining it would introduce target leakage and would produce performance figures that could not be reproduced in practice. All models in this paper therefore use only the 11 original clinical predictors, plus the three derived features introduced in the second stage.

3. Methods

3.1 Study design

The study consisted of two successive stages that share the same data and the same outcome definition (Figure 3). The baseline stage used a single stratified hold-out split, which gives a transparent estimate for a pipeline that is fitted once. The improvement stage replaced the hold-out split with stratified cross-validation, because a test set of 60 patients yields unstable estimates, and it enlarged both the feature set and the set of model families.

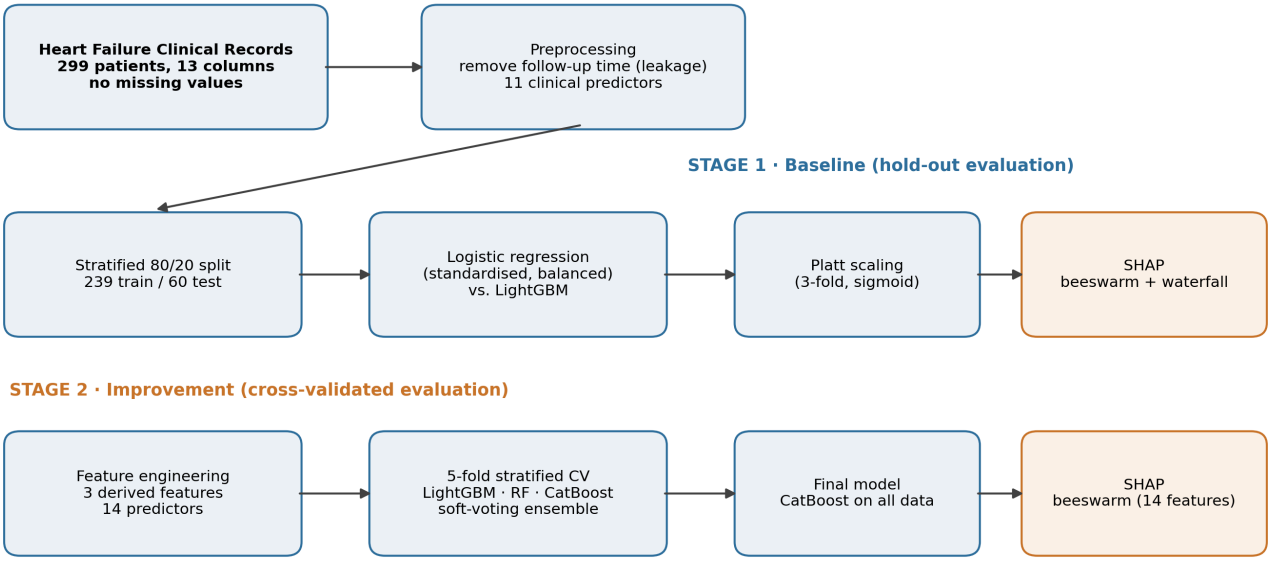


Figure 3. Overview of the study design: data and preprocessing, the baseline stage evaluated on a hold-out set, and the improvement stage evaluated by cross-validation.

3.2 Stage 1: baseline with hold-out evaluation

Data partitioning. The 11 predictors (all columns except `DEATH_EVENT` and `time`) and the outcome were split into a training set of 239 patients and a test set of 60 patients with a stratified 80/20 split and a fixed random seed (42), so that the proportion of deaths is preserved in both subsets. The test set was not used for any fitting or model-selection decision.

Logistic regression baseline. A logistic regression with balanced class weights was fitted after standardisation of the predictors to zero mean and unit variance, inside a single scikit-learn pipeline so that the scaling parameters were estimated on the training data only. Scaling is necessary for this model because the predictors differ by several orders of magnitude (platelets are of the order of 10^5, serum creatinine of the order of 1). Regularisation was left at the library default (L2 penalty, C = 1).

LightGBM classifier. A LightGBM gradient-boosted tree classifier (Ke et al., 2017) was fitted with 100 trees, a maximum depth of 3 and a learning rate of 0.05. The positive class was up-weighted by the ratio of negative to positive cases in the training set (approximately 2.1) through `scale_pos_weight`. The shallow trees and low learning rate were chosen in advance as a conservative setting for a small sample, and no search over hyperparameters was carried out.

Probability calibration. Class weighting improves sensitivity to the minority class but shifts the predicted probabilities away from empirical frequencies. The LightGBM model was therefore wrapped in `CalibratedClassifierCV` with sigmoid (Platt) scaling and

three internal folds fitted on the training set. Platt scaling maps the raw model score s to a calibrated probability through a logistic function whose two parameters, A and B, are estimated on held-out folds (Platt, 1999):

$$P(y = 1 \mid s) = \frac{1}{1 + \exp(A s + B)}$$

3.3 Stage 2: feature engineering, cross-validation and ensembling

Derived features. Three features were computed from the existing columns, bringing the predictor set from 11 to 14 variables:

- `renal_cardiac_ratio` = serum creatinine divided by ejection fraction, which combines a renal and a cardiac marker in a single quantity, so that a patient with both impaired renal function and reduced pumping capacity receives a high value;
- `creatinine_sodium_ratio` = serum creatinine divided by serum sodium;
- `age_hypertension` = age multiplied by the hypertension indicator, which equals the patient's age if hypertensive and zero otherwise.

None of these transformations estimates a parameter from the data, so computing them before the cross-validation split does not allow information from a validation fold to enter training.

Cross-validation. All patients were evaluated with 5-fold stratified cross-validation (shuffled, seed 42), so that each patient is used for validation exactly once and each fold contains about 60 patients with the outcome proportion preserved. In every fold each model was refitted from scratch on the four training folds and evaluated on the fifth. The positive-class weight was computed once from the full outcome vector (203/96, approximately 2.115) and not recomputed within each fold; it depends only on class counts and not on any predictor, so its influence on the estimates is negligible, although it is not strictly nested.

Models. Three model families were compared on the 14-feature set: LightGBM, a random forest (Breiman, 2001) and CatBoost (Prokhorenkova et al., 2018) (Table 3). A fourth model combined the three by soft voting, that is, by taking the unweighted mean of their predicted probabilities in each fold.

Model	Stage	Main settings
Logistic regression	1	Standardised inputs; L2 penalty (default); balanced class weights; seed 42
LightGBM	1	100 trees; maximum depth 3; learning rate 0.05; `scale_pos_weight` ≈ 2.1 (training set); seed 42

Model	Stage	Main settings
LightGBM, Platt-calibrated	1	Sigmoid calibration; 3 internal folds; fitted on the training set
LightGBM	2	60 trees; maximum depth 3; learning rate 0.03; `subsample` 0.8; `colsample_bytree` 0.8; `scale_pos_weight` ≈ 2.115; seed 42
Random forest	2	100 trees; maximum depth 4; balanced class weights; seed 42
CatBoost	2	100 iterations; depth 3; learning rate 0.04; `scale_pos_weight` ≈ 2.115; seed 42
Soft-voting ensemble	2	Unweighted mean of the probabilities of CatBoost, random forest and LightGBM

Table 3. Models and settings used in each stage. All settings were fixed by hand; no hyperparameter search was performed.

CatBoost was fitted on all 299 patients to obtain the final model that was explained and serialised.

3.4 Evaluation metrics

Discrimination was measured by the area under the receiver operating characteristic curve (ROC-AUC) and by the area under the precision-recall curve estimated as average precision (PR-AUC). The PR-AUC is emphasised, following Saito and Rehmsmeier (2015), because it responds to the performance on the minority positive class and because its no-skill value equals the prevalence (about 0.32), which makes its scale easy to interpret. Probability quality was measured by the Brier score (Brier, 1950), the mean squared difference between the predicted probability p and the observed outcome y over N patients, where lower values are better:

```
\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2
```

In the first stage all metrics were computed once on the 60 test patients. In the second stage they were computed in each fold and are reported as the mean over the five folds together with the standard deviation across folds (population form, with denominator 5). Accuracy is not used, since it is misleading under class imbalance. The performance targets for the project were a ROC-AUC of at least 0.85 and a Brier score of at most 0.15.

3.5 Explainability

Feature attributions were computed with SHAP (Lundberg and Lee, 2017) using the exact tree explainer of Lundberg et al. (2020). For a patient with feature vector x, the model output on the log-odds scale decomposes additively into a baseline value, which is the mean model output, and one contribution per feature:

```
f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j(x)
```

Positive values of φ_j raise the predicted risk and negative values lower it. In the first stage the explainer was applied to the fitted, uncalibrated LightGBM model and evaluated on the 60 test patients; global behaviour is summarised by a beeswarm plot, and a local explanation is given by a waterfall plot for the test patient with the highest predicted risk. In the second stage the explainer was applied to the final CatBoost model and evaluated on all 299 patients, and global behaviour is summarised by a beeswarm plot. Because the two models have different output scales, the magnitudes of the SHAP values are not comparable between stages.

3.6 Software and reproducibility

All computations were run in Python notebooks on a standard CPU, using pandas and NumPy for data handling, scikit-learn for splitting, preprocessing, logistic regression, random forest, calibration and metrics, LightGBM and CatBoost for gradient boosting, SHAP for attribution, matplotlib and seaborn for graphics, and joblib for serialisation. Every stochastic component was seeded with 42. The package versions were not recorded in the notebooks. The final models were serialised together with their ordered feature names as `clinical_lgbm_model.pkl` (stage 1, LightGBM, uncalibrated) and `final_catboost_clinical_model.pkl` (stage 2, CatBoost, uncalibrated, fitted on all patients).

4. Results

4.1 Stage 1: baseline comparison and calibration

On the 60 test patients, LightGBM achieved a higher ROC-AUC than the logistic regression baseline (0.759 against 0.734), whereas the baseline obtained the higher PR-AUC (0.591 against 0.542), as shown in Table 4 and Figure 4a.

Model	ROC-AUC	PR-AUC
Logistic regression (baseline)	0.734	0.591

Model	ROC-AUC	PR-AUC
LightGBM	0.759	0.542
LightGBM, calibrated (Platt)	0.741	0.511

Table 4. Discrimination on the hold-out test set (n = 60).

The two models are therefore not clearly separable. With roughly 20 deaths and 40 survivors in the test set there are about 800 pairs of one death and one survivor, and the ROC-AUC difference of 0.025 corresponds to the ordering of only about 20 of these pairs, so neither difference can be regarded as established. All three models are clearly above the no-skill PR-AUC of about 0.32, with PR-AUC values of 0.51 to 0.59, which shows that the predictors carry real signal. Calibration lowered both the ROC-AUC (0.759 to 0.741) and the PR-AUC (0.542 to 0.511). This is plausible, because the calibrated model is assembled from sub-models fitted on resampled parts of an already small training set, but the experiment does not isolate the cause.

Platt scaling lowered the Brier score of the LightGBM model from 0.1919 to 0.1862 (Table 5, Figure 4b), an absolute reduction of 0.006. The calibrated value is about 15% below the no-skill reference of 0.218, so the probabilities are informative, but it is above the target of 0.15. The Brier score of the logistic regression baseline was not computed in the notebook, so no calibration comparison between the two model families is available.

Model	Brier score
LightGBM, uncalibrated	0.1919
LightGBM, Platt-calibrated	0.1862
No-skill reference (prevalence)	≈ 0.218

Table 5. Brier score on the hold-out test set (lower is better).

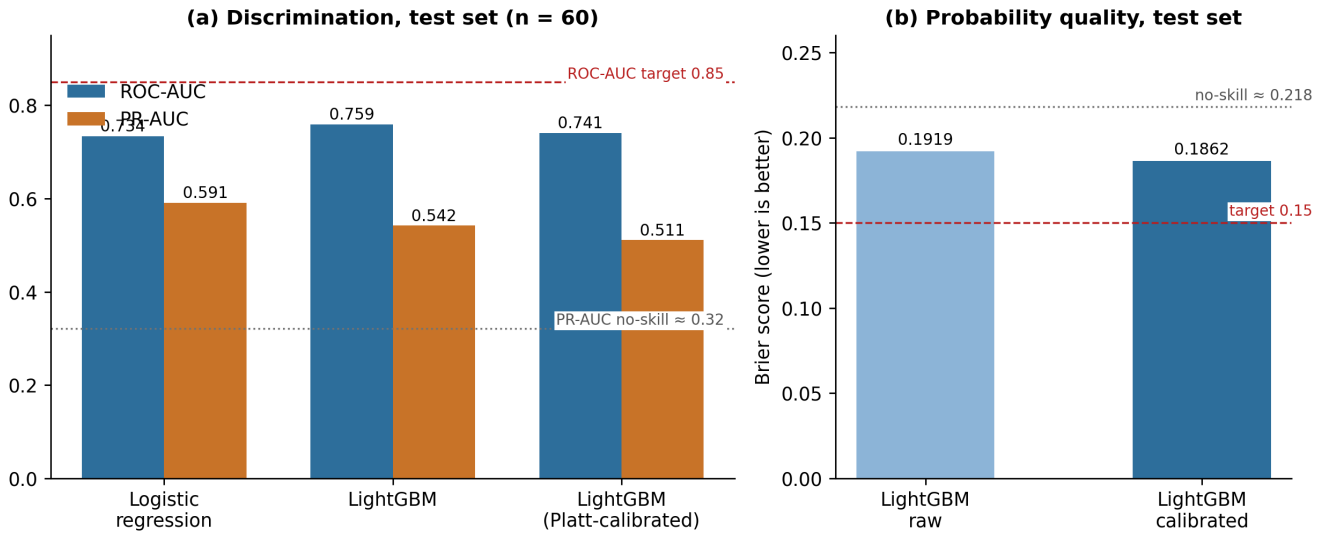


Figure 4. Stage 1 results on the test set (n = 60): (a) ROC-AUC and PR-AUC of the three models; (b) Brier score of LightGBM before and after Platt scaling. Dashed lines mark the project targets and dotted lines the no-skill references. Plotted from the values in Tables 4 and 5.

4.2 Stage 2: derived features

The three derived features have clear differences between survivors and non-survivors (Table 6, Figure 5). The ratio of serum creatinine to ejection fraction has a correlation of +0.395 with the outcome, higher than that of any of the original predictors, the largest of which is serum creatinine at +0.294 (Figure 6). The ratio of creatinine to sodium (+0.299) is comparable to serum creatinine alone, and the age-hypertension product (+0.112) is only weakly associated with the outcome.

Feature	Mean (SD)	Median	Maximum	Mean, survived	Mean, died	Correlation with outcome
renal_cardiac_ratio	0.0409 (0.0324)	0.0300	0.2686	0.0322	0.0596	+0.395
creatinine_sodium_ratio	0.0103 (0.0077)	0.0080	0.0707	0.0087	0.0136	+0.299
age_hypertension	21.89 (30.57)	0	95	19.55	26.84	+0.112

Table 6. Summary statistics of the three derived features, group means and correlation with DEATH_EVENT (N = 299).

The median of `renal_cardiac_ratio` is about 0.027 among survivors and about 0.05 among patients who died, and the interquartile ranges of the two groups overlap only narrowly. The distribution is right-skewed, with a maximum of 0.269 against a median of 0.030. For `age_hypertension` the median is zero in both groups, because 65% of patients

are not hypertensive, and the separation lies in the upper quartile (about 51 among survivors and about 63 among deaths). These are univariate associations between variables that are themselves derived from strongly related inputs, and they do not show that the new features add information beyond the original ones.

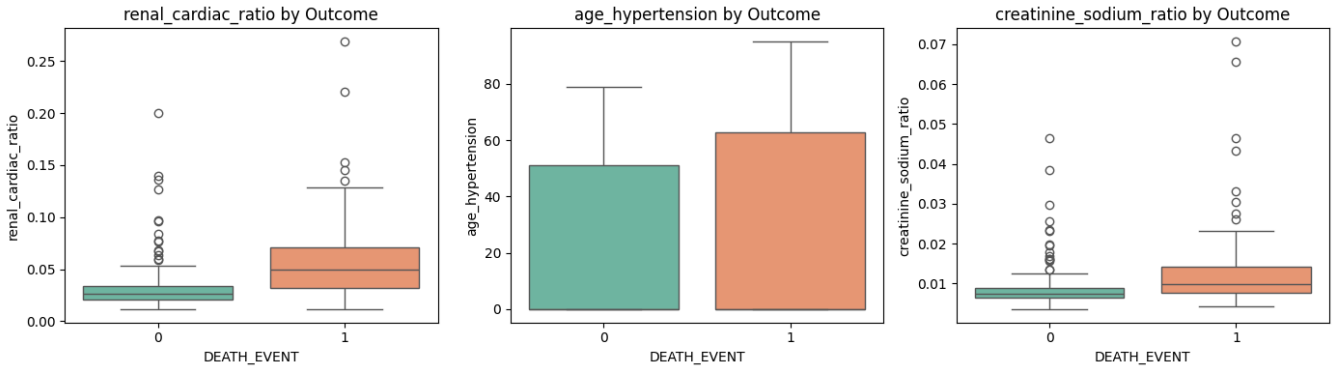


Figure 5. Box plots of the three derived features for survivors (0) and patients who died (1).

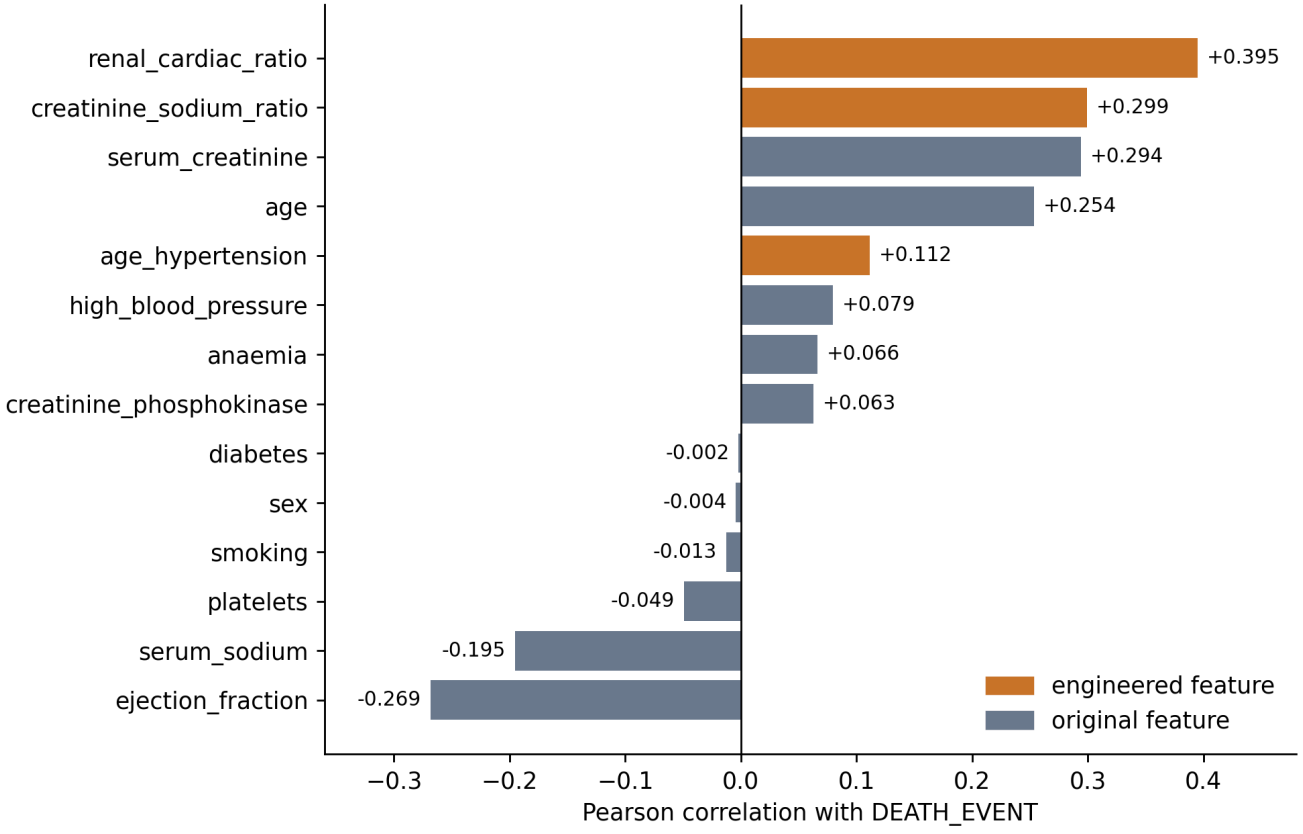


Figure 6. Correlation of the 14 predictors with DEATH_EVENT. Derived features are shown in orange. The correlation of `renal_cardiac_ratio` exceeds that of every original variable.

4.3 Stage 2: cross-validated comparison of models

The four models are very close under 5-fold cross-validation (Table 7, Figure 7). The mean ROC-AUC ranges from 0.784 (LightGBM) to 0.798 (CatBoost) and the mean PR-AUC from 0.662 (LightGBM) to 0.676 (CatBoost), while the Brier scores lie between 0.174 and 0.178.

Model	ROC-AUC	PR-AUC	Brier score
LightGBM	0.784 (0.075)	0.662 (0.113)	0.175
Random forest	0.795 (0.073)	0.664 (0.112)	0.174
CatBoost	0.798 (0.060)	0.676 (0.118)	0.178
Soft-voting ensemble	0.797 (0.071)	0.668 (0.129)	0.174

Table 7. 5-fold stratified cross-validation on the 14-feature set: mean (standard deviation across folds). Brier scores are means over folds for uncalibrated probabilities.

The standard deviation across folds is 0.060 to 0.075 for ROC-AUC and 0.11 to 0.13 for PR-AUC, which is roughly four to nine times the spread between the model means (0.014 for both measures). Every difference between models is therefore much smaller than the variation between folds, and the models cannot be ranked with confidence. CatBoost has the highest mean ROC-AUC and PR-AUC and the smallest ROC-AUC variability, and the random forest and the ensemble have the lowest Brier score (0.174), but these margins are not meaningful given the fold-to-fold variability. The ensemble did not improve on its best member in ROC-AUC (0.797 against 0.798) or in PR-AUC (0.668 against 0.676), and its Brier score is lower than CatBoost's by only 0.004.

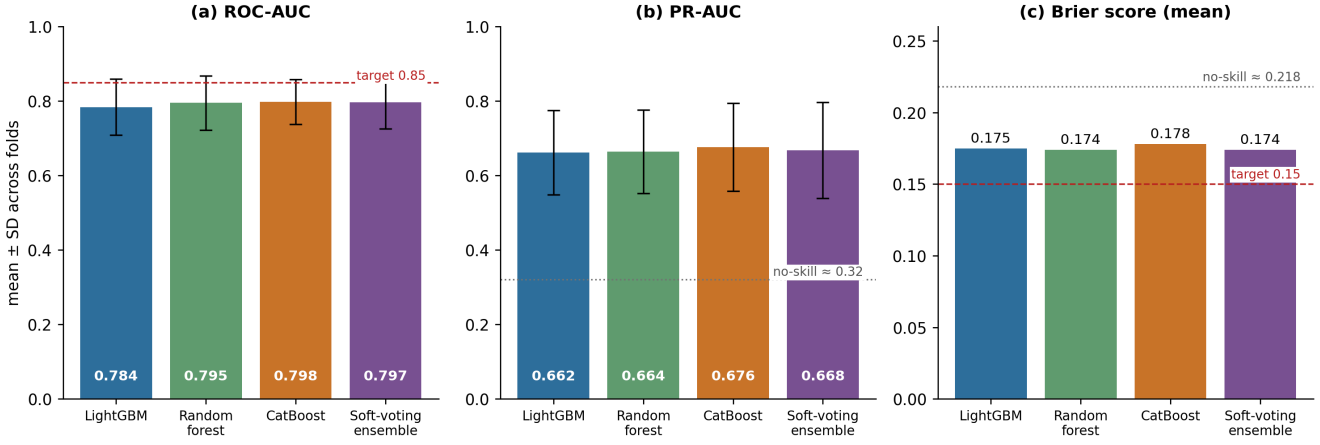


Figure 7. Stage 2 cross-validated results: (a) ROC-AUC and (b) PR-AUC, mean with standard deviation across folds; (c) mean Brier score. Dashed lines mark the project targets and dotted lines the no-skill references. Plotted from the values in Table 7.

The best mean ROC-AUC (0.798) is 0.052 below the target of 0.85, and the best Brier score (0.174) is 0.024 above the target of 0.15 and about 20% below the no-skill reference. None of the four models satisfied either target. For LightGBM, the mean ROC-AUC rose from 0.759 on the stage 1 hold-out set to 0.784 under cross-validation, the PR-AUC from

0.542 to 0.662, and the Brier score from 0.192 to 0.175. These figures are encouraging, but they cannot be attributed to the derived features alone, because the evaluation protocol, the feature set and the hyperparameters all changed between the two stages.

5. Explainability results

5.1 Global attribution in stage 1 (LightGBM)

The beeswarm plot on the left of Figure 8 ranks the predictors of the stage 1 LightGBM model by mean absolute SHAP value as follows: serum creatinine, ejection fraction, age, creatinine phosphokinase, platelets, high blood pressure, serum sodium, anaemia, diabetes, smoking and sex. Each point is one test patient, its horizontal position is the contribution of that feature to the patient's log-odds of death, and its colour is the value of the feature.

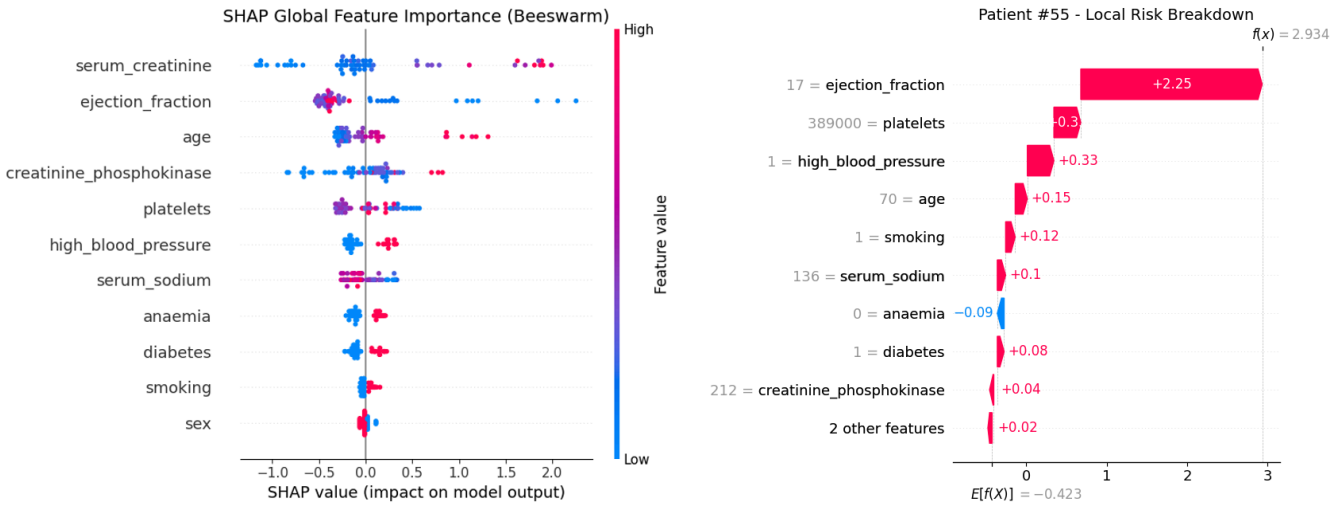


Figure 8. Stage 1 SHAP explanation of the LightGBM model on the 60 test patients. Left: beeswarm plot of global feature attributions. Right: waterfall plot of the local explanation for the test patient with the highest predicted risk (position 55 in the test set). Values are in log-odds.

The direction of the main effects agrees with the exploratory analysis and with clinical expectation:

- A high serum creatinine, which reflects impaired renal function, raises the predicted risk, and low values lower it. This variable has the largest individual contributions, from roughly −1.2 to +2.0 log-odds.
- A low ejection fraction raises the risk, and a small group of patients with very low values receives large positive contributions of up to about +2.3; high values lower the risk by about 0.4.
- Higher age raises the risk, with the strongest effects concentrated in the oldest patients (up to about +1.3).

- Lower serum sodium tends to raise the risk, in line with the group means in Table 2.
- The presence of hypertension, anaemia, diabetes or smoking contributes a small positive amount, and sex contributes almost nothing.
- Creatinine phosphokinase and platelets show non-monotone patterns, with both low and high values appearing on either side of zero, which suggests that their effect is interactive or dominated by noise at this sample size.

5.2 Local attribution for a high-risk patient

The waterfall plot on the right of Figure 8 shows how the score of a single patient is assembled. The patient is the one with the highest predicted risk in the test set. The mean model output over the test set is −0.42 log-odds, and the prediction for this patient is +2.93 log-odds, which corresponds to a raw, uncalibrated probability of about 0.95. The individual contributions are listed in Table 8.

Feature	Patient value	Contribution (log-odds)
ejection_fraction	17%	+2.25
platelets	389,000	approximately +0.3
high_blood_pressure	1	+0.33
age	70	+0.15
smoking	1	+0.12
serum_sodium	136	+0.10
anaemia	0	−0.09
diabetes	1	+0.08
creatinine_phosphokinase	212	+0.04
serum_creatinine and sex (combined)	not shown	+0.02

Table 8. SHAP contributions for the highest-risk test patient (baseline −0.42, prediction +2.93 log-odds).

An ejection fraction of 17% accounts for 2.25 of the total shift of about 3.36 log-odds above the baseline, roughly two thirds of the explanation. The remaining contributions are small and mostly positive, and the absence of anaemia reduces the score slightly. Serum creatinine and sex together contribute only +0.02, so for this patient the elevated risk is explained almost entirely by cardiac function and not by renal markers. This example

shows the intended use of the method in a decision-support setting: the same patient-level breakdown can be shown with the risk score, so that a clinician can see which measurements are responsible for it and judge whether the explanation is plausible.

5.3 Global attribution in stage 2 (CatBoost)

For the final CatBoost model, evaluated on all 299 patients, the 14 predictors are ranked as follows (Figure 9): `renal_cardiac_ratio`, age, ejection fraction, `creatinine_sodium_ratio`, `age_hypertension`, serum sodium, serum creatinine, creatinine phosphokinase, platelets, high blood pressure, anaemia, diabetes, sex and smoking.

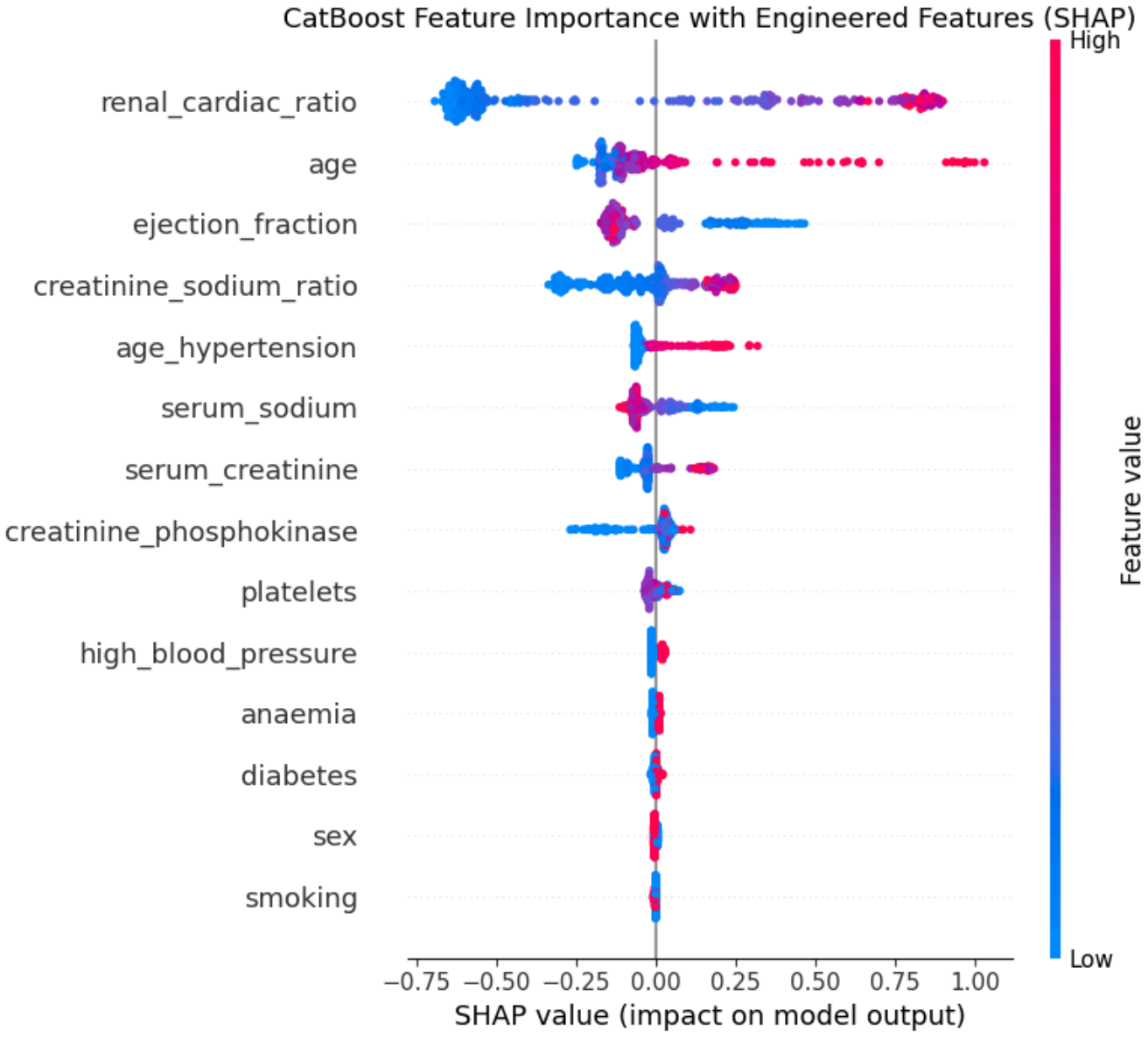


Figure 9. Stage 2 SHAP beeswarm plot of the final CatBoost model on all 299 patients, including the derived features. Values are in log-odds.

- The derived feature `renal_cardiac_ratio` is the dominant predictor. High values raise the predicted risk by up to roughly +0.9 log-odds and low values lower it by about 0.65, which is a wider range than any other variable.
- Age is second. High values contribute positively, and the oldest patients reach about +1.0, while young patients contribute about −0.2.
- A low ejection fraction raises the risk (up to about +0.45), and high values lower it slightly (about −0.15).
- A high `creatinine_sodium_ratio` raises the risk (up to about +0.25) and low values lower it (down to about −0.3). A high `age_hypertension`, that is an older hypertensive patient, contributes up to about +0.3, and a value of zero lowers the risk slightly (about −0.1).
- Lower serum sodium raises the risk (by up to about +0.25), and higher sodium lowers it slightly (about −0.1).
- Serum creatinine itself has a small effect (about −0.1 to +0.15) once the ratios that contain it are in the model, and low creatinine phosphokinase values lower the risk by up to about 0.25. Platelets and the binary indicators, including sex and smoking, contribute very little.

5.4 Comparison of the two explanations

Rank	Stage 1: LightGBM (test set, 11 features)	Stage 2: CatBoost (all patients, 14 features)
1	serum_creatinine	renal_cardiac_ratio
2	ejection_fraction	age
3	age	ejection_fraction
4	creatinine_phosphokinase	creatinine_sodium_ratio
5	platelets	age_hypertension
6	high_blood_pressure	serum_sodium
7	serum_sodium	serum_creatinine
8	anaemia	creatinine_phosphokinase
9	diabetes	platelets
10	smoking	high_blood_pressure
11	sex	anaemia
12	not applicable	diabetes

Rank	Stage 1: LightGBM (test set, 11 features)	Stage 2: CatBoost (all patients, 14 features)
13	not applicable	sex
14	not applicable	smoking

Table 9. Ranking of the predictors by mean absolute SHAP value in the two stages.

Both explanations agree on the clinical story: renal function, left-ventricular ejection fraction and age are the principal determinants of predicted risk, serum sodium has a smaller effect in the expected direction, and sex, smoking, diabetes and anaemia contribute little. The main change is that serum creatinine, which was the leading variable in stage 1, ranks seventh in stage 2. This does not mean that renal function became less important. Its information is now shared with the two derived features that contain it, and SHAP divides credit among correlated variables, so the rank of an individual variable cannot be compared directly between models with different feature sets. The magnitudes of the SHAP values are likewise not comparable, since the two models produce outputs on different scales.

6. Discussion

6.1 Principal findings

Three findings stand out. First, a small set of routine measurements gives moderate discrimination for mortality in heart failure: across both stages and all models the ROC-AUC lay between 0.73 and 0.80, clearly above chance, and the PR-AUC lay between 0.51 and 0.68, about 1.6 to 2.1 times the prevalence of 0.32. Second, the choice of model family had little visible effect. LightGBM, random forest, CatBoost and their average gave practically the same cross-validated scores, and the logistic regression baseline was within a few hundredths of LightGBM on the hold-out set. Third, the explanations are stable and clinically coherent across models and stages: renal function, ejection fraction and age drive the predicted risk, and a ratio that combines creatinine and ejection fraction is the single most informative feature that was constructed.

6.2 Performance relative to the targets

Requirement	Target	Stage 1 (hold-out)	Stage 2 (cross-validation)	Status
ROC-AUC	≥ 0.85	0.759 (LightGBM); 0.741 calibrated	0.798 (CatBoost)	Not met
PR-AUC	Reported	0.542; 0.511 calibrated	0.676 (CatBoost)	Reported
Brier score	≤ 0.15	0.186 (calibrated)	0.174 (random forest, ensemble)	Not met

Requirement	Target	Stage 1 (hold-out)	Stage 2 (cross-validation)	Status
Platt calibration	Applied	Applied	Not applied	Partly met
Global and local SHAP	Required	Beeswarm and waterfall	Beeswarm	Partly met
Hyperparameter tuning	Optuna	Not done	Not done	Not done
Serialised model	Required	`clinical_lgbm_model.pkl`	`final_catboost_clinical_model.pkl`	Met
REST service, container, latency ≤ 15 ms	Required	Not implemented	Not implemented	Not done

Table 10. Status of the project targets and deliverables after both stages.

Neither ROC-AUC nor Brier score reached its target. The best cross-validated ROC-AUC (0.798) is 0.052 below the target of 0.85, and the best Brier score (0.174) is 0.024 above 0.15. The main reason is probably the size of the cohort. The dataset has 299 patients, fewer than the range of 1,000 to 100,000 records contemplated in the project specification, and 96 events. With so few events a flexible model cannot learn stable interactions, and the estimate of its performance is itself uncertain, as the fold-to-fold standard deviation of 0.06 to 0.075 in ROC-AUC shows. A second reason is the exclusion of follow-up time, which is the most strongly correlated variable ($r = -0.53$) but cannot be used in a prediction made at the bedside. An analysis that kept it would be expected to show higher apparent discrimination, but the figure could not be reproduced in practice. It is therefore reasonable to regard the targets as set for a larger dataset, and not as evidence of a modeling failure.

6.3 Feature engineering, model family and ensembling

The derived ratio of creatinine to ejection fraction has a stronger univariate association with the outcome than any original variable, and it is the leading feature in the SHAP ranking of the final model. This agrees with the finding of Chicco and Jurman (2020) that these two measurements carry most of the predictive information, and it suggests that expressing their joint effect explicitly is helpful when the sample is too small for a tree ensemble to discover the interaction by itself. The present data do not allow the contribution of the derived features to be separated from other changes between the stages, and an ablation on identical folds is needed to quantify it. The lack of any benefit from ensembling is not surprising: the three members have similar inductive biases, are fitted to the same 14 features and are all strongly regularised, so their errors are highly correlated and averaging them removes little variance.

6.4 Calibration

The Brier scores of 0.174 to 0.186 are 15% to 20% below the no-skill reference of 0.218, so the predicted probabilities carry information, but they remain above the target. Part of the shortfall is likely caused by the positive-class weighting, which inflates predicted probabilities by design. Platt scaling gave only a small gain in stage 1 (0.192 to 0.186), and it was not applied in stage 2, so the second-stage Brier scores are those of uncalibrated, class-weighted models. Sigmoid calibration assumes a particular monotone distortion and is preferred when the calibration set is small, whereas isotonic regression needs more data (Niculescu-Mizil and Caruana, 2005). Training without class weights, or calibrating the final model on out-of-fold predictions, are options that were not tested. Because no calibration curve was drawn, any statement about calibration rests on a single summary measure.

6.5 Clinical reading of the explanations

The attributions agree with well-recognised prognostic markers in heart failure. A raised serum creatinine reflects reduced renal function, which commonly accompanies a failing circulation, a low ejection fraction reflects reduced systolic function, higher age increases baseline risk, and a low serum sodium is a recognised marker of advanced disease. That the model reproduces this pattern without being told it is evidence that it has learned structure and not artefacts of the sample. The local explanation of Figure 8 illustrates the practical value of the method: for the highest-risk patient the score was driven almost entirely by an ejection fraction of 17%, information that a clinician can verify immediately. At the same time, SHAP values describe the behaviour of the fitted model and not causal effects, and a variable with a small attribution, such as smoking or diabetes, should not be read as clinically unimportant.

7. Limitations

Sample size and uncertainty. The cohort contains 299 patients and 96 events, and the stage 1 test set only 60 patients. At this size a single reordered pair of patients changes the ROC-AUC by about 0.001, and the sampling uncertainty of every figure in this paper is large. No confidence intervals were computed, and the cross-validation standard deviations show that differences of the order of 0.01 to 0.03 between models are within noise.

Evaluation design. The two stages differ in protocol, feature set and hyperparameters, so they are not directly comparable, and the rise of the LightGBM ROC-AUC from 0.759 to 0.784 cannot be attributed to a single change. No untouched final test set was kept in stage 2, so all stage 2 figures come from the same data that guided the choice of models and settings.

Hyperparameter selection. No systematic search (for example with Optuna) was carried out in either stage. The second-stage settings were fixed by hand and differ from those of the first stage (for LightGBM, 60 trees at a learning rate of 0.03 against 100 trees at 0.05). The notebooks do not record how they were chosen, so selection on the same folds used for reporting cannot be excluded, and it would bias the cross-validated scores upward. Nested cross-validation or a final held-out test set would remove this risk. In addition, the row subsampling requested for LightGBM (`subsample` = 0.8) has no effect unless `subsample_freq` is set above zero, which it was not.

Calibration. Platt scaling was applied only in stage 1, calibration curves were not plotted, and the Brier score of the logistic regression baseline was not computed.

Explainability. In stage 2 the final CatBoost model was explained on the same 299 patients it was fitted on, so the attributions describe the fitted model and are not a held-out assessment. In stage 1 the explainer used only 60 patients, which makes the rankings of the weaker predictors noisy. SHAP divides credit among correlated variables, so ranks cannot be compared across feature sets.

Data and outcome. The data come from a single institute and a single period, and no external validation was performed, so generalisation to other populations is unknown. The outcome is treated as a binary indicator of death during a follow-up of 4 to 285 days, which ignores the timing of events and the censoring of patients who left follow-up early. A time-to-event model would use follow-up information in a legitimate way and could give a more appropriate risk estimate.

Deployment. The REST service, containerisation and latency measurement required by the project were not implemented, so the inference-latency requirement of 15 ms remains untested.

8. Conclusion

This study built and evaluated a leakage-free mortality-risk model for patients with heart failure from 11 routine clinical measurements and three derived features, in two stages. In the baseline stage, LightGBM achieved a test ROC-AUC of 0.759 against 0.734 for a standardised logistic regression, the logistic regression had the higher PR-AUC (0.591 against 0.542), and Platt scaling lowered the Brier score of LightGBM from 0.192 to 0.186. In the improvement stage, LightGBM, random forest, CatBoost and a soft-voting ensemble reached mean cross-validated ROC-AUC values of 0.784 to 0.798, PR-AUC values of 0.662 to 0.676 and Brier scores of 0.174 to 0.178. These differences were smaller than the variation between folds, so no model could be shown to be better than another. CatBoost, which had the highest mean ROC-AUC and PR-AUC, was retained as the final model.

SHAP attribution gave consistent and clinically plausible explanations in both stages. Serum creatinine, ejection fraction and age were the leading predictors of the baseline model, and the derived ratio of serum creatinine to ejection fraction, age and ejection fraction were the leading predictors of the final model. The patient-level waterfall explanation showed how the score for an individual patient can be decomposed and checked against clinical knowledge. The pre-specified targets of ROC-AUC ⩾ 0.85 and Brier score ⩽ 0.15 were not reached, and the principal explanation is the small cohort of 299 patients.

The results support the view that compact, shallow tree models with explicit explanations are a reasonable approach to mortality-risk stratification from routine data, but they do not support a claim of clinical readiness. The next steps are a controlled ablation of the derived features on identical folds with bootstrap confidence intervals, systematic hyperparameter tuning within nested cross-validation, calibration of the final model with calibration curves, held-out local explanations, validation on an external cohort, a time-to-event formulation that uses follow-up information, and implementation of the inference service with measured latency.

Data and code availability

The dataset is publicly available on Kaggle as the Heart Failure Clinical Data collection (andrewmvd/heart-failure-clinical-data). The analysis was carried out in two notebooks, one for each stage, and the serialised models are `clinical_lgbm_model.pkl` and `final_catboost_clinical_model.pkl`.

References

Ahmad, T., Munir, A., Bhatti, S. H., Aftab, M., and Raza, M. A. (2017). Survival analysis of heart failure patients: a case study. *PLoS ONE*, 12(7), e0181001.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.

Chicco, D., and Jurman, G. (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, 20, 16.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., and Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2, 56–67.

Lundberg, S. M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

Niculescu-Mizil, A., and Caruana, R. (2005). Predicting good probabilities with supervised learning. *Proceedings of the 22nd International Conference on Machine Learning*, 625–632.

Pedregosa, F., et al. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31.

Saito, T., and Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3), e0118432.